

EX02: Control of the automatic nervous system under alcohol

The objective of this exercise is to analyse the control of the autonomic nervous system at rest and after alcohol consumption using breathing, mean blood pressure and interbeat signals.

```
In [1]: import numpy as np
import pylab as py
py.ion()
py.close('all')
import scipy.signal as sp
import m03_ex2_ext as my_plot
```

Load signals of a subject at rest.

```
In [2]: x = np.genfromtxt('heart_1.dat', delimiter=' ').T
x = {'rr':x[0], 'bp':x[1], 'resp':x[2]}
# Load signals of a subject after alcohol consumption.
y = np.genfromtxt('heart_2.dat', delimiter=' ').T
y = {'rr':y[0], 'bp':y[1], 'resp':y[2]}
# Signals are sampled at 4 Hz.
fs = 4
# Generate the time for the recordings.
t = np.arange(len(x['rr']))/fs
```

Cardiac interbeats, mean blood pressure and respiration volume of a subject at rest.

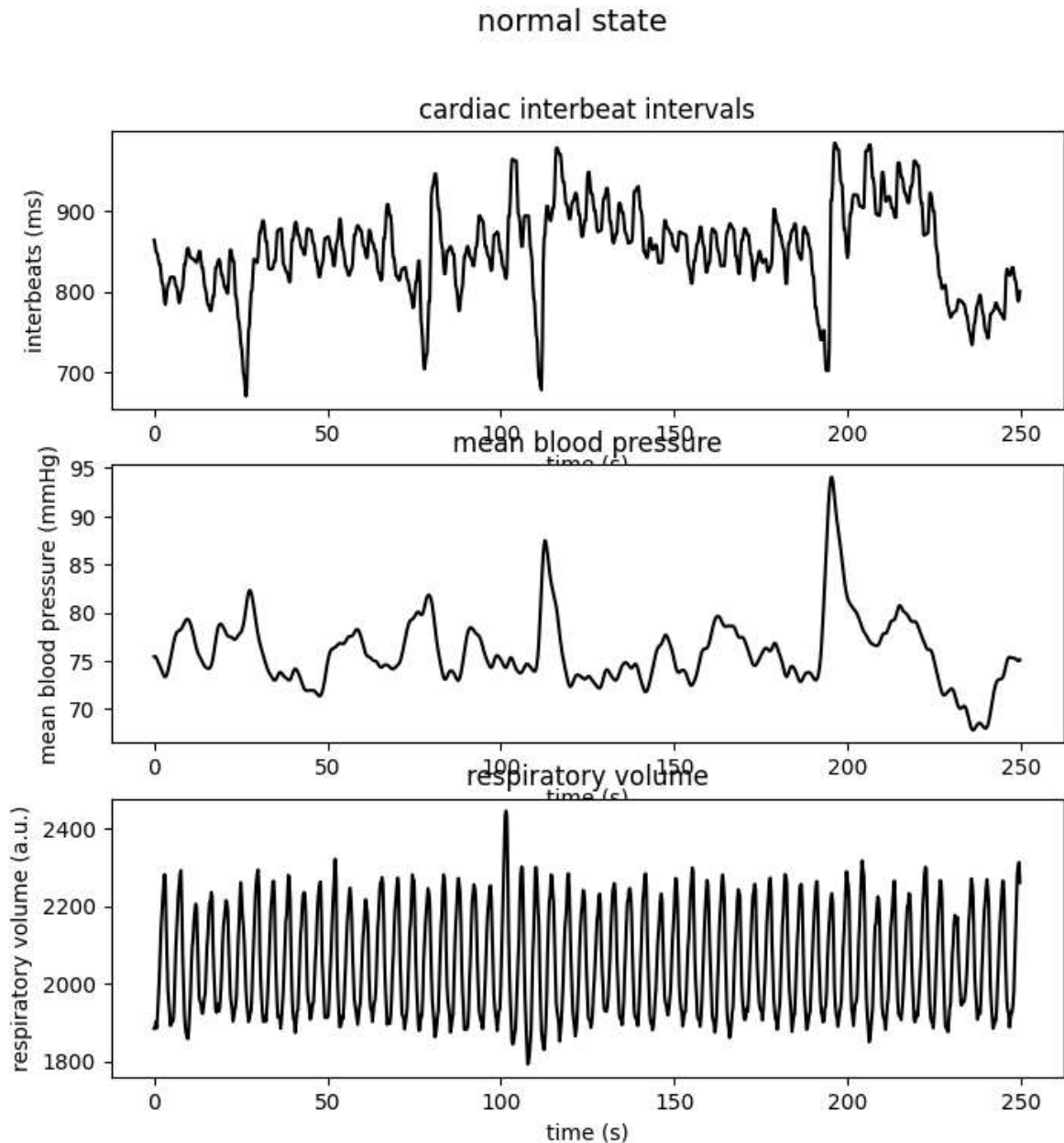
Q: Comment the different signals and their relationships.

The respiratory signal presents a high-frequency oscillation. The cardiac interbeat interval follows this rhythm but with smaller fluctuations. The mean blood pressure varies more slowly, showing broader oscillations that are partly influenced by both respiration and heart rate changes. Together, the signals illustrate the combined effects of breathing on heart rhythm and blood pressure regulation.

Q: Which signals are related and how?

The respiratory signal and the cardiac interbeat interval are directly coupled through respiration, while blood pressure and the cardiac interbeat interval are inversely linked, the cardiac interbeats with high amplitude arise at the same time of peaks of high blood pressure. The heart try to regulate blood pressure and making it the more constant possible

```
In [3]: my_plot.plot_time(x, t, 'normal state')
```



Cardiac interbeats, mean blood pressure and respiration volume of a subject after alcohol consumption.

Q: Comment the different signals and their relationships.

After alcohol consumption, the respiratory signal remains regular, but the cardiac interbeat interval shows smaller and less clear fluctuations compared to rest. The mean blood pressure varies more irregularly, with slower oscillations and weaker synchronization with respiration. The overall coupling between the signals appears reduced.

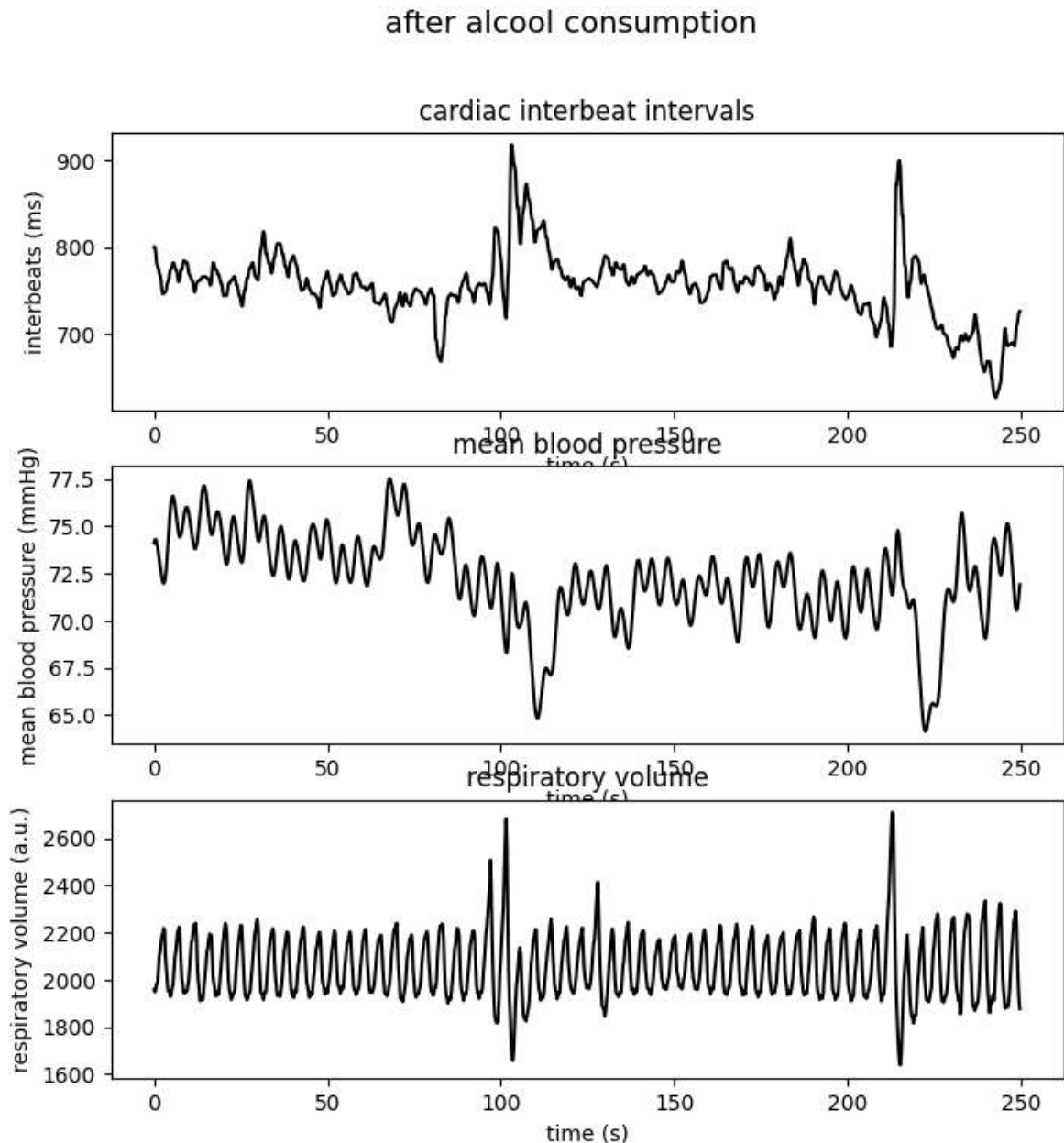
Q: Which signals are related and how?

Respiration still influences the cardiac interbeat interval, but the link is weaker than with no alcohol consumption. Blood pressure and the cardiac interbeat interval remain connected through the reflex that stabilizes pressure, yet this interaction is less efficient and less visible.

Q: What are the differences with rest recording of previous figure?

Compared to sober condition, the cardiac interbeat interval after alcohol consumption exhibits reduced variability and weaker modulations driven by respiration. Blood pressure also becomes less regular and more variable. In summary, alcohol weakens the relationship between respiration, heart rhythm, and blood pressure.

```
In [4]: my_plot.plot_time(y, t, 'after alchool consumption')
```



Compute the intercorrelation of the signals of the subject at rest.

Q: Comment the oscillation present in the different signals.

The breathing autocorrelation plot show a symetric linearly decading signal. Wich is kind of normal as the respiratory signal act like a sinusoide with almost same amplitude leading to a non random signal. The cardiac interbeat show a slow frequency signal with little high frequency modulation. The fast component is due to the constant regulation of the blood pressure while the slow component are due to extreme case of high/low blood pressure. The blood pressure autocorrelation just show slow fluctuations that are due to change of blood pressure

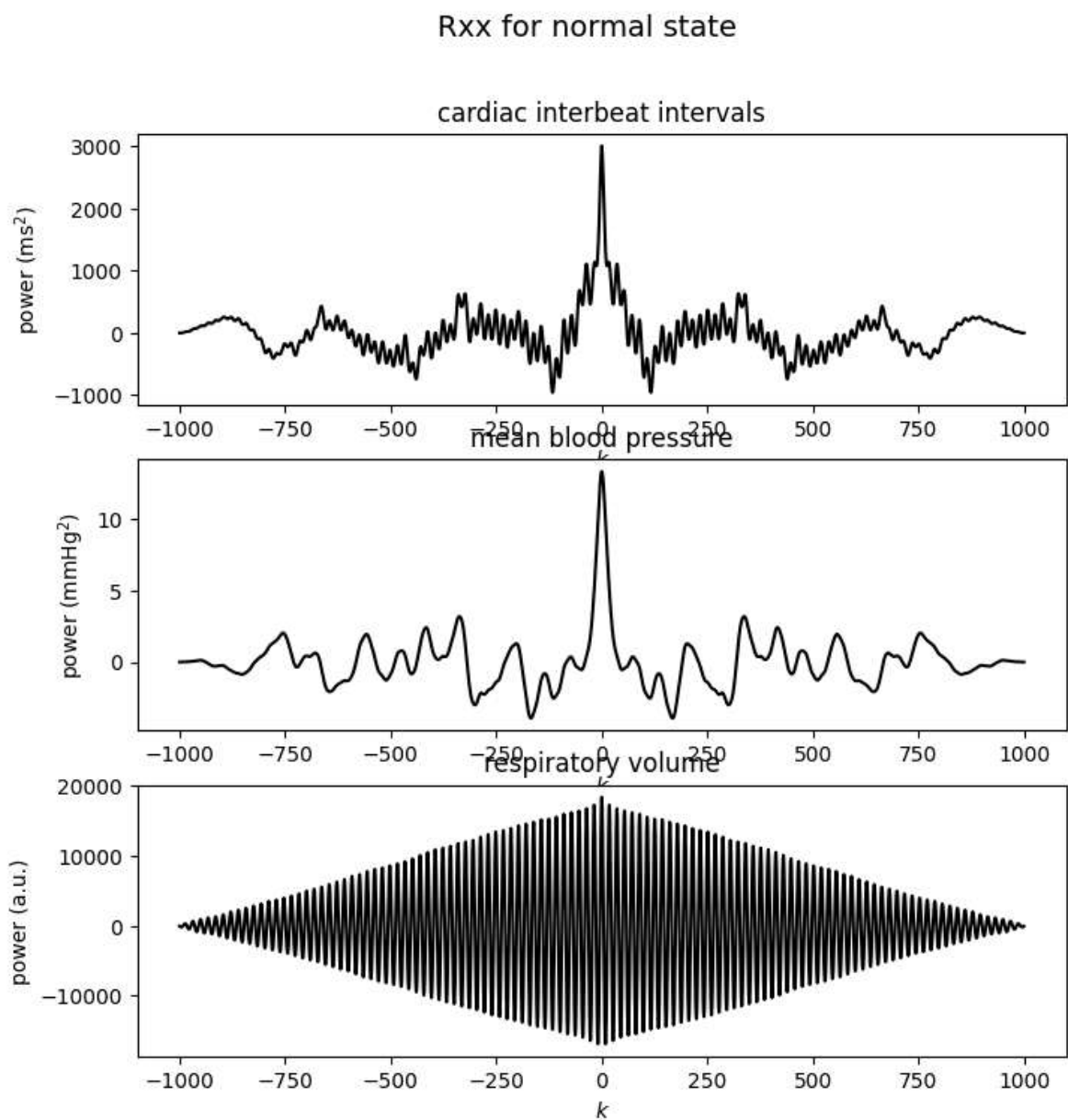
Q: Which signals are related and how.

The cardiac interbeat signal is correlated with the mean blood pressure signals as explained above

```
In [5]: def my_corr(x):  
        rxx = np.correlate(x-np.mean(x), x-np.mean(x), mode='full')/len(x)  
        return rxx
```

```
In [6]: x['rxx_rr'] = my_corr(x['rr'])  
        x['rxx_bp'] = my_corr(x['bp'])  
        x['rxx_resp'] = my_corr(x['resp'])
```

```
In [7]: my_plot.plot_rxx(x, 'Rxx for normal state')
```



Compute the intercorrelation of the signals of the subject after alcohol consumption.

Q: Comment the oscillation present in the different signals.

Respiration shows a very strong and regular oscillation, visible as repeated peaks in the autocorrelation. The interbeat intervals present a central peak with smaller side oscillations, indicating irregular variability with some periodic components. Mean blood pressure displays a central peak with superimposed fast oscillations, reflecting smaller and less stable rhythmicity.

Q: Which signals are related and how.

Respiration continues to influence interbeat intervals and blood pressure, but the coupling appears weaker. Heart rate variations with breathing are attenuated, and blood pressure oscillations follow respiration with reduced amplitude. The inverse relationship between blood pressure and interbeat intervals is still present but less evident.

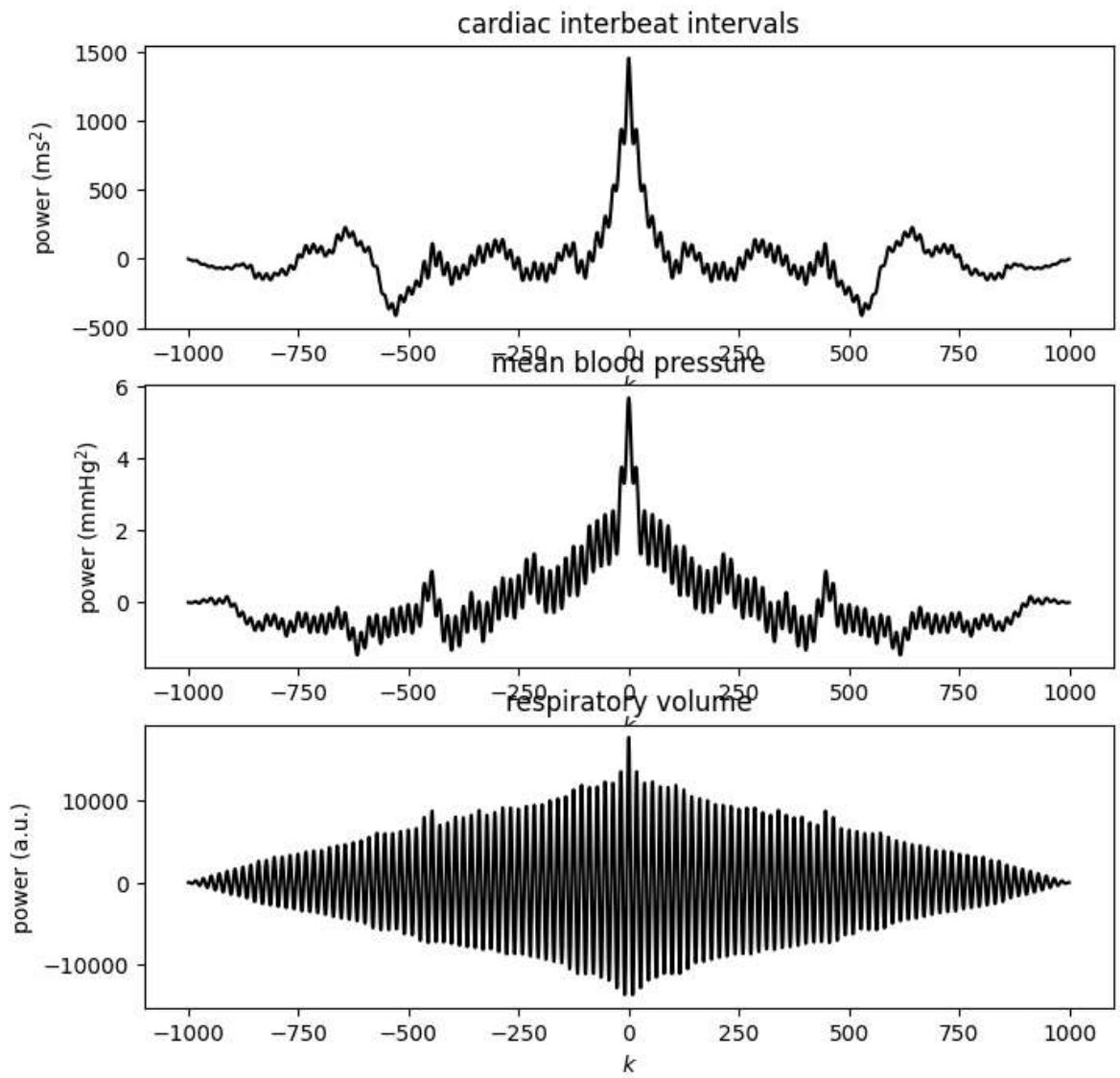
Q: What difference do you observe with the previous figure?

Compared with the resting state, the autocorrelations after alcohol show reduced amplitude and less regular oscillatory patterns, especially in interbeat intervals and mean blood pressure. This indicates a weakening of the autonomic modulation, with respiration remaining regular but its influence on cardiovascular signals diminished.

```
In [8]: y['rxx_rr'] = my_corr(y['rr'])  
y['rxx_bp'] = my_corr(y['bp'])  
y['rxx_resp'] = my_corr(y['resp'])
```

```
In [9]: my_plot.plot_rxx(y, 'Rxx after alcohol consumption')
```

Rxx after alcohol consumption



Compute the PSD of the signal for the subject at rest.

Q: How the different peaks are related to the control of the autonomic nervous system?

The sharp peak of respiration (0.20–0.25 Hz, HF band) marks respiratory sinus arrhythmia and vagal (parasympathetic) modulation; the LF peak in cardiac intervals and blood pressure (~0.1 Hz) reflects baroreflex-driven oscillations (“Mayer waves”) with mixed sympathetic–parasympathetic influence; the very-low-frequency background in blood pressure (<0.04 Hz) reflects slower regulatory mechanisms.

Q: Do the positions and amplitude of the peaks confirm you previous findings;

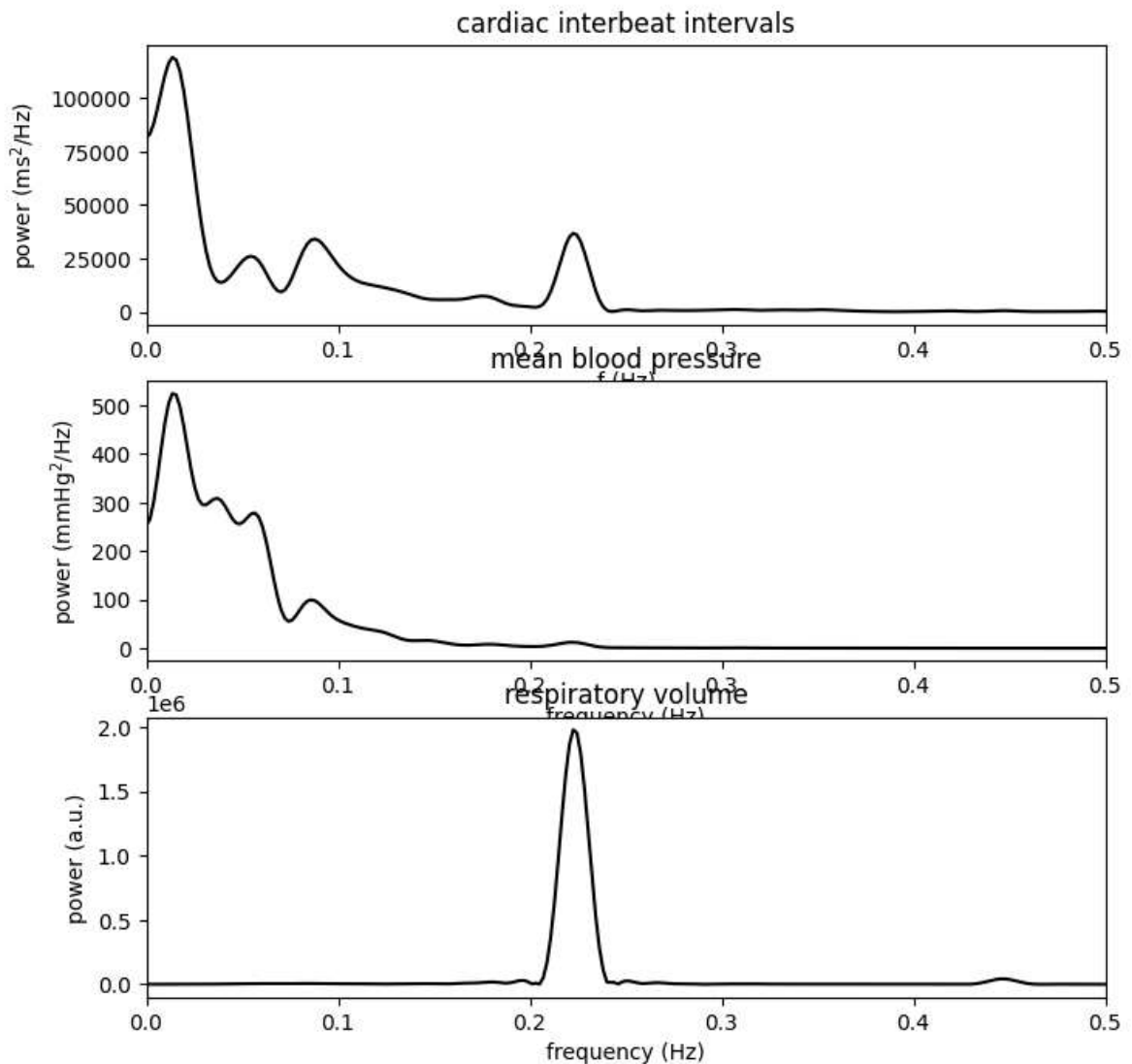
Yes. The high-frequency peak around the respiratory rate is strong in respiration and also visible in the interbeat intervals, confirming the coupling between breathing and heart rate. In mean blood pressure this peak is weaker, while the spectrum is dominated by low-frequency components, which agrees with the observation of slower and less regular oscillations in blood pressure and its inverse relation with interbeat intervals at rest.

```
In [19]: from scipy.signal import windows
def my_psd(x, half_win=250):
    interval = np.arange(-half_win, half_win+1)+len(x)//2
    x_sub = x[interval]
    psd = np.abs(np.fft.fft(windows.hann(len(interval))*x_sub, 2048))
    return psd
```

```
In [20]: x['RR'] = my_psd(x['rxx_rr'])
x['BP'] = my_psd(x['rxx_bp'])
x['RESP'] = my_psd(x['rxx_resp'])
```

```
In [21]: my_plot.plot_X(x, fs, 'PSD for normal state')
```

PSD for normal state



Compute the PSD of the signal for the subject after alcohol consumption.

Q: How the different peaks are related to the control of the autonomic nervous system?

The spectrum shows a clear respiratory peak around 0.2 Hz in the breathing signal, while in the interbeat intervals and blood pressure this peak is smaller, reflecting a weaker modulation of cardiovascular activity by respiration. Low-frequency components are still visible, mainly in blood pressure, indicating ongoing baroreflex and sympathetic activity.

Q: Do the positions and amplitude of the peaks confirm you previous findings;

Yes. The respiratory peak appears consistently at the expected frequency, confirming the link between breathing and autonomic modulation. Its reduced amplitude in interbeat intervals and blood pressure supports the observation that autonomic influence on these signals is less pronounced, while blood pressure continues to display stronger low-frequency power.


```
In [ ]: y['RR'] = my_psd(y['rxx_rr'])
        y['BP'] = my_psd(y['rxx_bp'])
        y['RESP'] = my_psd(y['rxx_resp'])
```

```
In [ ]: my_plot.plot_X(y, fs, 'PSD after alchool consumption')
```

Plot the PSDs of the signals for the two conditions.

Q: Discuss the differences.

Breathing peaks at the same frequency (~0.22–0.25 Hz) in both conditions, with only a slight reduction in amplitude after alcohol. In the cardiac interbeat intervals, the high-frequency peak at the respiratory rate is strong at rest but largely suppressed after alcohol, indicating weaker respiratory sinus arrhythmia. Mean blood pressure shows reduced low-frequency power (<0.15 Hz) and a more evident high-frequency peak at the respiratory rate. Overall, spectral power in cardiovascular signals is lower after alcohol, especially in the LF and HF components, pointing to diminished autonomic modulation, with respiration remaining regular but exerting less influence on heart rate and blood pressure.

```
In [ ]: my_plot.plot_XY(x, y, fs, 'Comparison of the PSD')
```